

A Clustering Approach to Triage Categorization in A&E Departments

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Abstract—UK accident and emergency departments have to treat patients from arrival to discharge, transfer or admission within four hours. Scientific methods to determine the triage category are important for the hospital to effectively manage resources and staff. In this paper, two clustering algorithms, i.e., K -means and fuzzy c-means clustering, are applied to establish the triage category via data collected by the accident and emergency (A&E) departments. Silhouette coefficients are then employed to evaluate the clustering validity. Finally, the proposed methods are tested on a data set provided by a hospital group in West London. The experimental results show that the established triage category fits the data very well.

I. INTRODUCTION

The past few decades have witnessed the increasing research interest in the National Health Service (NHS). Comparing to other medical services, one of the key challenges of the NHS is that the NHS works for 24 hours a day, 7 days a week and any patient requiring urgent treatment can go to NHS urgent care at any time. However, due to the unlimited, unscheduled emergency cases, the NHS is suffering from the increasing financial pressure [6]. Within the NHS, the A&E departments are one of the most important units, which deal with genuine life-threatening emergencies. Usually, the A&E departments are required to ensure that at least 98% of patients are treated from arrival to discharge, transfer or admission within 4 hours.

In order to meet the 4-hour requirement in A&E departments, considerable research efforts have been deployed on data analysis in the A&E departments. For instance, the expectation maximization algorithm has been used in [24] for modeling the throughput of emergency departments according to time-series data. In [10], a discrete-event simulation model has been presented to describe the throughput of A&E departments. In [23], the clustering analysis technique has been applied to allocate hospital bed resources while, in [11], the same technique has been applied to support the process of building the simulation modeling of obstetrics and gynecological patient flows in hospital. In [7], an intelligent patient management system has been proposed based on clustering techniques and discrete-event simulation to facilitate improvements in patient care. In [14], classification techniques are used to improve the validity of classifying the general practice cases

in A&E departments. In [21], with the help of classification techniques, bed overflow has been studied. It should be pointed out that an appropriate triage category plays an important role in the management of medical resources and can effectively improve treatment efficiency. Indeed, with the accurate triage category, it enables staff to treat the patients with serious injuries first and reallocate the staff to the departments which is in urgent need of manpower. Therefore, it is important to generate an appropriate triage category based on data collected by the A&E departments.

Actually, the triage category can be generated by the clustering analysis technique. Clustering analysis aims to divide data into meaningful or useful groups. K -means clustering is a commonly used clustering approach which has been successfully employed in a variety of fields such as biology, signal processing computer vision, market segmentation, and healthcare [1], [3], [8], [20], [22]. K -means clustering is actually a hard clustering approach where data points can only belong to one cluster. In contrast, fuzzy c-means (FCM) clustering approach allows data points to belong to several clusters simultaneously with different degrees of membership and is capable of dealing with some special cases. For example, in [3], the gene expression data has been studied by using FCM clustering algorithm. In [1], a novel FCM clustering based algorithm has been presented for fuzzy segmentation of magnetic resonance imaging data and estimation of intensity inhomogeneities. It is worth mentioning that, no matter whether there is a natural cluster structure in a data set, almost every clustering algorithm can always find clusters in the given data set. Therefore, it is necessary to evaluate the clustering performance after the category is generated by the chosen clustering approach. Our main motivation in this paper is to utilize the K -means clustering and FCM clustering approaches with the cluster evaluation method to generate an appropriate triage category.

Motivated by the above discussion, we investigate the generation problem of the triage category in the A&E departments with the hope of improving the healthcare management. First, the data is pre-processed to eliminate irrelevant, redundant information and missing data. Then, by using the K -means clustering and FCM clustering techniques, the triage category in A&E departments is generated. Subsequently, the generated triage category is evaluated by the silhouette cluster evaluation

method. The experimental results show that the obtained triage category fits the data very well.

The remaining part of this paper is organized in the following manner. First of all, two clustering algorithms, which are K -means and FCM clustering algorithms, are reviewed in Section II. In Section III, the experimental analysis is discussed in detail. Conclusions and future topics of research are presented in Section IV.

II. K -MEANS CLUSTERING AND FUZZY C-MEANS CLUSTERING

Clustering analysis aims to discover the natural structure of a set of data points. Moreover, data clustering has been used to study the degree of the similarity of the data [12], [13]. Clustering methods can be mainly categorized into two types: hierarchical and partitional. Here we only discuss about two well-known partitional algorithms, K -means and FCM clustering algorithms.

A. K -means Clustering

As a traditional partitional clustering algorithm, K -means was independently presented in different research areas [16], [17]. K -means algorithm is used to partition the data points so as to minimize the squared error between the empirical mean of a cluster and the points in the cluster. The main steps of K -means algorithm are illustrated in Algorithm 1.

Algorithm 1 Basic K -means algorithm

1. Select K points as initial cluster centroids.
 2. Assign each point to its closest cluster centroid.
 3. Recalculate the centroids of each cluster when all points have been assigned.
 4. Repeat Steps 2 and 3 until the centroids do not change.
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The K -means algorithm starts with selecting K random initial points as the cluster centroids. The clustering performance is affected due to different initializations as the algorithm picks K random cluster centroids each time we run the program. As a result, the experiment should be repeated several times to guarantee clustering performance. Data points are assigned to their closest cluster according to the distances between the data points to each cluster centroid. Points assigned to the same centroid form a new cluster. The centroid of each cluster is then updated based on these new points assigned to the cluster. The process of recalculating the centroids and reassignment of data points stops when a stopping criterion is met. For instance, we can stop the algorithm when the centroids remain the same. Taking computation time into consideration, we can use another stopping criterion by calculating the difference between the objective function in two iterations. If the difference is smaller than the threshold value, the algorithm will stop. The K -means clustering algorithm can be viewed as a gradient descent algorithm and the detailed proof of convergence of the algorithm has been shown in [4].

B. Fuzzy c -means Clustering

FCM clustering algorithm, an extension of the basic K -means algorithm, was first introduced by Dunn [9] and later improved by Bezdek in 1981 [5]. In K -means clustering, each data point is assigned to a single cluster. However, it might be more appropriate to assign the data points to a cluster with a membership value what represents the degree of membership that each point belongs to each single cluster, especially for the clusters which are not well-separated. In FCM clustering algorithm, each data point is partitioned based on the degree of membership that each data point belongs to the cluster. The main steps of FCM clustering algorithm are presented in Algorithm 2.

Algorithm 2 Basic Fuzzy c -means algorithm

1. Randomly initialize the cluster membership values.
 2. Calculate the centroid of each cluster using the membership values.
 3. Update the membership values based on the cluster centroids.
 4. Repeat Steps 2 and 3 until the centroids don't change.
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The usage of fuzzy partition in FCM clustering brings the challenge of adjusting fuzzy membership values comparing to the basic K -means algorithm. In FCM, the membership values are randomly initialised and subject to the constraint that the sum of membership values of any data point must be 1. The cluster centroids are computed based on the initialized membership values. Next, the membership values are updated on the basis of new cluster centroids. Detailed calculation of the membership value and cluster centroids are given in [5]. By iteratively updating the cluster centroids and the membership values for each data point, the new centroids move to the location which indicates that the objective function is minimized. It should be noted that the objective function represents the distance from any data point to a cluster centroid weighted by the data point's membership value. The FCM clustering algorithm stops when the improvement of the objective function is less than the given threshold or the centroids remain the same.

III. EXPERIMENTAL ANALYSIS

The introduced methods are tested on the data set provided by a hospital in West London which includes the A&E Department, the Urgent Care Centre, and the Minor Injury Unit. The data represents the A&E attendances and emergency admissions from 2014 to 2015. The total numbers of patients attending the A&E Department, the Urgent Care Centre and the Minor Injury Unit are 51,713, 15,151 and 60,122, respectively. The total number of incidents recorded in these three emergency departments is 126,986.

A. Data pre-processing

In our data, each row represents a patient's record, and each column represents an attribute with respect to the incident. The data includes 25 attributes which are the patient NHS

number, provider code, provider name, arrival date time, attendance conclusion date time, date time seen for treatment, arrival month, year month, fiscal year label, core healthcare resource group (HRG), HRG description, mode of arrival, mode of arrival description, referral source, referral source description, A&E department description, general practitioner (GP) practice code, patient age, clinical commissioning groups (CCGs), first diagnosis, diagnosis description, attendance disposal, attendance disposal description, and postcode sector of usual address, respectively.

At this stage, the data should be pre-processed in order to remove noise, detect outliers and normalize some columns of data. The pre-processing process will facilitate later development of the data analysis. The raw data includes some nulls and missing values, and hence we remove the records with missing data. In our data, we delete the 3778 incidents because the patients' records on the treatment time are missing. We also need to identify certain attributes that are of interest to us. In our data, we have some redundant attributes which are used to describe previous attributes, e.g., mode of arrival and mode of arrival description. By applying statistical analysis and computing the Pearson correlation coefficient, some attributes like CCGs and postcode sector of usual address are not useful for our research on the triage category.

It should be mentioned that the provided data is recorded at actual time such as arrival date time, attendance conclusion date time, and date time seen for treatment. To calculate the treatment time, we compute the time interval between the attendance conclusion date time and date time seen for treatment. It's worth noting that we unify the measurement scale of the data by labeling the attributes individually. Particularly, each attribute has different number of labels, for example, we have 37 diagnosis, 2 modes of arrival and 49 classes of treatment time. If we directly apply clustering algorithms to the raw data without any modification, the results will be highly influenced by the attributes with larger number of classes. Hence we normalize the data according to the following equation:

$$X_{\text{Norm}} = \frac{x - \min(X)}{\max(X) - \min(X)} \quad (1)$$

where X_{Norm} represents the normalized value, $\min(X)$ represents the minimum value of data X , $\max(X)$ describes the maximum value of data X .

B. Parameter Setting

In this subsection, detailed information on parameter setting is addressed. For K -means clustering algorithm, the inputs are data X and the number of clusters is K . The pre-processed data is imported as a matrix to MATLAB. A crucial first step in K -means clustering algorithm is to choose an appropriate K . However, so far, there has been no criterion of the selection of K , we choose K heuristically. Hence K is tested from 2 to 20 iteratively, on the basis of the literature [2]. Moreover, the distance metric and cluster initialization are defined by the user in K -means clustering algorithm. We here randomly

choose K cluster centroids among the existing data points. However, sometimes the randomly chosen clusters may be very close to each other which may lead to unsatisfactory results. As a result, we add an additional option to repeat the experiment 5 times for each of K . It should be noted that there have been plenty of distance metrics, and here we adopt the four commonly used metrics which are squared Euclidean distance (sqeuclidean), cityblock distance, correlation distance and cosine distance. Mathematical expressions of the four distance metrics are given in [20]. Experimental results for the usage of the distance metrics are further discussed in III-C.

Similarly, in FCM clustering algorithm, we have to import the data matrix and choose the number of clusters K . The selection of K in FCM clustering algorithm is the same as that in K -means clustering algorithm. Additionally, the parameter setting of distance metric, the maximum iteration, and cluster initialization are the same as those in K -means clustering algorithm. Note that the fuzzy partition matrix U which indicates the degree of membership of the data point in the cluster is randomly initialized. Note that the membership value for a data point will sum to one. The clustering process stops when the maximum number of iterations is reached, or when the objective function improvement between two consecutive iterations is less than the given threshold. The maximum iteration and the threshold are set as 100 and $1e^{-5}$ in both K -means and FCM clustering algorithms.

C. Cluster Evaluation

Cluster evaluation is an important part of any cluster analysis. Through cluster evaluation, we can

- 1) determine if there exists any non-random structures in the data;
- 2) determine the appropriate number of clusters;
- 3) evaluate the performance of clustering algorithms without reference to external information; and
- 4) determine the clustering performance by comparing the obtained results to some externally known results such as the class labels provided by experts.

In general, there exist three types of evaluation methods to judge various aspects of cluster validity: 1) unsupervised cluster evaluation, 2) supervised cluster evaluation, 3) and relative cluster evaluation. In our work, the unsupervised cluster evaluation approach is adopted to evaluate the clustering performance. The measures that the unsupervised cluster evaluation approach uses include cohesion and separation. The cohesion measure determines how the data points in a cluster are closely related and the separation measure reveals how a cluster is distinct from others.

In [18], the silhouette coefficient approach which combines both cohesion and separation is chosen in this paper to evaluate the clustering performance. The value of the silhouette coefficient varies between -1 and 1. The positive values show that the data points are very far from neighbouring clusters, and negative values represent that the data points are probably assigned to the wrong cluster. If the silhouette coefficient is 0, it means the data point lies equally far away from two

clusters and can be assigned to any one of them. By taking the average of all the silhouette coefficients of data points, the average silhouette coefficients can be obtained. Generally, the overall clustering performance is evaluated by the average silhouette coefficients.

D. Results and Discussion

In this subsection, we present the experimental results with the different number of clusters and the distance metric. Figs. 1 and 2 show, respectively, the average silhouette coefficients in different number of clusters for K -means and FCM clustering algorithms using sqeuclidean distance, cityblock distance, correlation distance and cosine distance. The computation times for the K -means and FCM clustering algorithms are given in Figs. 3 and 4, respectively. Figs. 5 and 6 present, respectively, the silhouette cluster evaluations for K -means and FCM clustering algorithms where the number of clusters is taken as 2 and the correlation distance is selected.

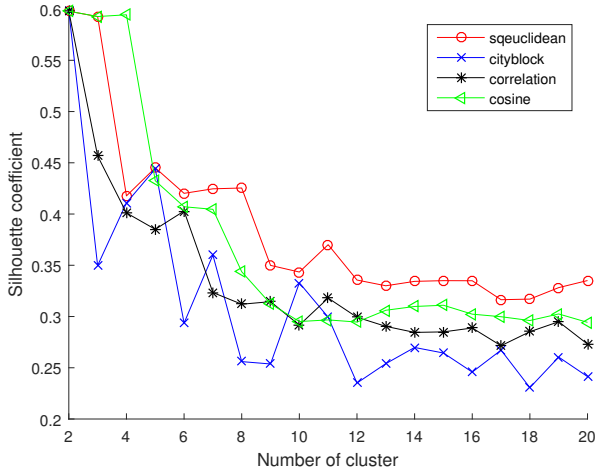


Fig. 1: Average silhouette coefficients for K -means clustering using sqeuclidean distance, cityblock distance, correlation distance and cosine distance

In Fig. 1, we can see a obvious downward trend of the average silhouette coefficients for K -means clustering algorithm as K varies from two to twenty for all four distance metrics. All the maximum average silhouette values for the four distance metrics are 0.5979. The average silhouette coefficients using sqeuclidean distance metric is larger than ones using other three distance metrics. For the FCM clustering algorithm, it can be seen from Fig. 2 that only the average silhouette coefficients using correlation distance and cosine distance can reach the maximum value. Moreover, the average silhouette coefficients using correlation distance and cosine distance are exactly same. Generally, the higher silhouette coefficient means the better clustering performance. Therefore, it can be seen from both Figs. 1 and 2 that the optimal number of clusters is two for our data.

It can be seen from Fig. 3 that the K -means clustering algorithm by using cityblock distance consumes the most

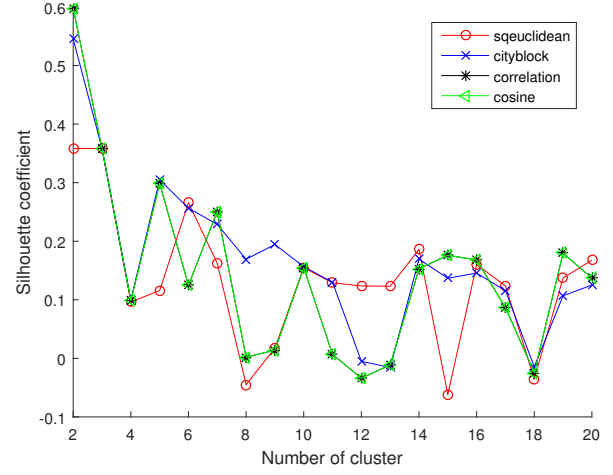


Fig. 2: Average silhouette coefficients for FCM clustering using sqeuclidean distance, cityblock distance, correlation distance and cosine distance

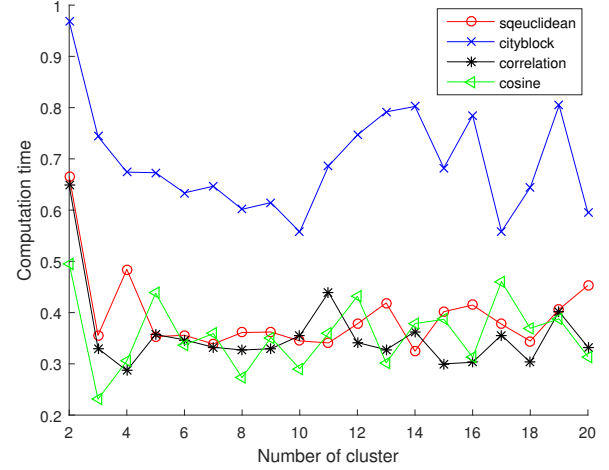


Fig. 3: Computation time for K -means clustering algorithm using sqeuclidean distance, cityblock distance, correlation distance and cosine distance

computation time. For the FCM clustering algorithm, it can be seen from that Fig. 4 that the computation time is almost linear with respect to the number of clusters except for few special points. Comparing to the K -means clustering algorithm, the FCM clustering algorithm consumes more computation time.

In Figs. 5 and 6, the number of clusters is taken as two and the correlation distance is used. It can be seen from Fig. 5 that all the silhouette coefficients of the K -means clustering algorithm are positive which indicates that all data points are assigned to the proper clusters. The same results for the FCM clustering algorithm are verified in Fig. 6.

From the above experiments, it can be seen that two clusters is appropriate and, based on it, the triage categories can be obtained which consists of serious and critical patients.

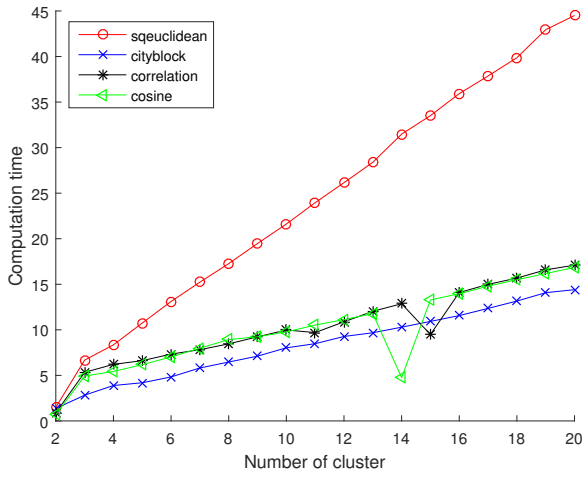


Fig. 4: Computation time for FCM clustering algorithm using sqeuclidean distance, cityblock distance, correlation distance and cosine distance

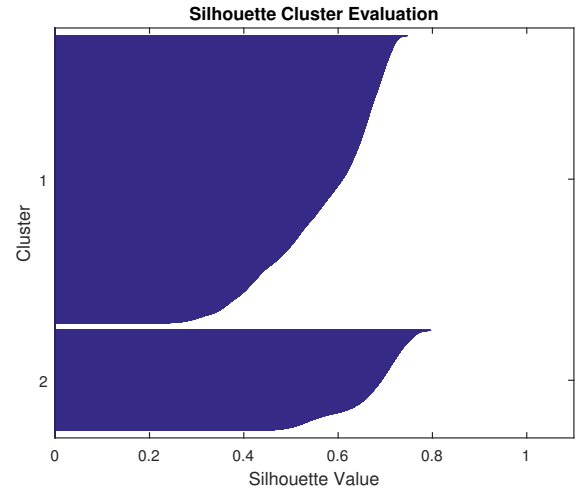


Fig. 6: Silhouette cluster evaluation for FCM clustering using correlation distance with two clusters

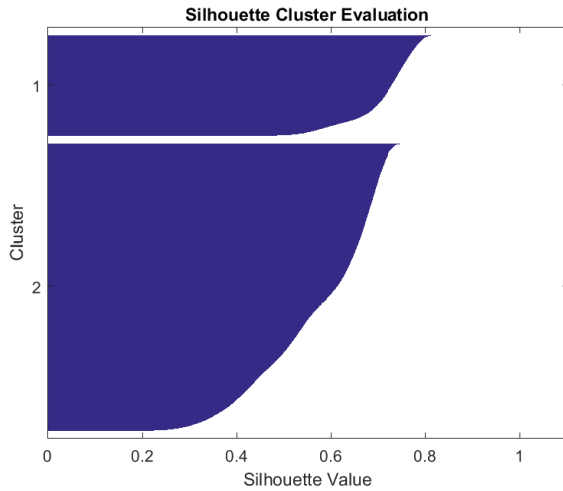


Fig. 5: Silhouette cluster evaluation for K -means clustering using correlation distance with two clusters

IV. CONCLUSION

In this paper, a new triage category in NHS A&E department has been established using K -means and FCM clustering algorithms. The results show that it is most appropriate to categorize the data into two groups. In this case, we divide the triage category into serious and critical patients. Our results are helpful for further analysis of the data from A&E departments.

In the future study of A&E data, three major research directions will be taken into consideration.

- Firstly, advanced clustering algorithms will be applied. Moreover, some optimization algorithms like the genetic algorithm and the particle swarm algorithm will be adopted to improve the clustering performance [19], [25].
- Secondly, since the A&E data includes some time-dependent attributes, the relationship between time and

such attributes can be modelled as a time series by using statistical tools. Further analysis on the data such as prediction is taken into consideration. More specifically, we can predict the usage requirements of the medical apparatus such as the X-ray, A&E beds and the B ultrasound.

- Thirdly, according to the outcome of the clustering algorithm, we can carry out data classification using deep neural networks [15], [26]. In addition, we are able to classify different departments into very busy, busy and normal according to date and time. On the basis of the arrival date, time and the patients' first diagnosis, we can classify the patients' waiting time into very long, long and short.

REFERENCES

- [1] M. N. Ahmed, S. M. Yamany, N. Mohamed, A. A. Farag and T. Moriarty, A modified fuzzy c-means algorithm for bias field estimation and segmentation of MRI data, *IEEE transactions on medical imaging*, vol. 21, no. 3, pp. 193–199, 2002.
- [2] S. P. Baker, B. o'Neill, Jr. W. Haddon and W. B. Long, The injury severity score: a method for describing patients with multiple injuries and evaluating emergency care, *Journal of Trauma and Acute Care Surgery*, vol. 14, no. 3, pp. 187–196, 1974.
- [3] A. Ben-Dor, R. Shamir and Z. Yakhini, Clustering gene expression patterns, *Journal of computational biology*, vol. 6, no. 3-4, pp. 281–297, 1999.
- [4] L. Bottou and Y. Bengio, Convergence properties of the k-means algorithms, *Advances in neural information processing systems*, pp. 585–592, 1995.
- [5] J. C. Bezdek, *Pattern Recognition with Fuzzy Objective Function Algorithms*, Plenum Press: New York, 1981.
- [6] Centre for Change and Innovation, Decide to Admit v Admit to Decide - Working Together to Improve Unscheduled Care in NHS Scotland, *National Conference Unplanned Care for Medical Conditions*, 2004.
- [7] A. Codrington-Virtue, T. Chausalet, P. Millard, P. Whittlestone and J. Kelly, A system for patient management based discrete-event simulation and hierarchical clustering, *19th IEEE Symposium on Computer-Based Medical Systems*, pp. 800–804, 2006.
- [8] I. S. Dhillon and D. S. Modha, Concept decompositions for large sparse text data using clustering, *Machine learning*, vol. 42, no. 1, pp. 143–175, 2001.

- [9] J. C. Dunn, A fuzzy relative of the ISODATA process and its use in detecting compact well-separated clusters, *Journal of Cybernetics*, vol. 3, no. 3, pp. 32–57, 1973.
- [10] J. Eatock, M. Clarke, C. Picton and T. Young, Meeting the four-hour deadline in an A&E department, *Journal of health organization and management*, vol. 25, no. 6, pp. 606–624, 2011.
- [11] M. W. Isken and B. Rajagopalan, Data mining to support simulation modeling of patient flow in hospitals, *Journal of medical systems*, col. 26, no. 2, pp. 179–197, 2002.
- [12] A. K. Jain, Data clustering: 50 years beyond K-means, *Pattern recognition letters*, vol. 31, no. 8, pp. 651–666, 2010.
- [13] A. K. Jain, M. N. Murty and P. J. Flynn, Data clustering: a review, *ACM computing surveys (CSUR)*, vol. 31, no. 3, pp. 264–323, 1999.
- [14] A. Lee, C. B. Hazlett, S. Chow, F. L. Lau, C. W. Kam, P. Wong and T. W. Wong, How to minimize inappropriate utilization of Accident and Emergency Departments: improve the validity of classifying the general practice cases amongst the A&E attendees, *Health Policy*, vol. 66, no. 2, pp. 159–168, 2003.
- [15] W. Liu, Z. Wang, X. Liu, N. Zeng, Y. Liu, and F. E. Alsaadi, A survey of deep neural network architectures and their applications, *Neurocomputing*, vol. 234, pp. 11–26, 2017.
- [16] S. Lloyd, Least squares quantization in PCM, *IEEE transactions on information theory*, vol. 28, no. 2, pp. 129–137, 1982.
- [17] J. MacQueen, Some methods for classification and analysis of multivariate observations. In *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability*, vol. 1, no. 14, pp. 281–297, 1967.
- [18] P. J. Rousseeuw, Silhouettes: a graphical aid to the interpretation and validation of cluster analysis, *Journal of computational and applied mathematics*, vol. 20, pp. 53–65, 1987.
- [19] B. Song, Z. Wang and L. Zou, On global smooth path planning for mobile robots using a novel multimodal delayed PSO algorithm, *Cognitive Computation*, vol. 9, no. 1, pp. 5–17, 2017.
- [20] P. N. Tan, M. Steinbach and V. Kumar, *Introduction to Data Mining*, Addison Wesley: Boston, 2005.
- [21] K. L. Teow, E. El-Darzi, C. Foo, X. Jin and J. Sim, Intelligent analysis of acute bed overflow in a tertiary hospital in Singapore, *Journal of medical systems*, vol. 36, no. 3, pp. 1873–1882, 2012.
- [22] R. W. Tothill, A. V. Tinker, J. George, R. Brown, S. B. Fox, S. Lade, D. S. Johnson, M. K. Trivett, D. Etemadmoghadam, B. Locandro and N. Traficante, Novel molecular subtypes of serous and endometrioid ovarian cancer linked to clinical outcome, *Clinical cancer research*, vol. 14, no. 16, pp. 5198–5208, 2008.
- [23] S. Walczak, W. E. Pofahl and R. J. Scorpio, A decision support tool for allocating hospital bed resources and determining required acuity of care, *Decision support systems*, vol. 34, no. 4, pp. 445–456, 2003.
- [24] Z. Wang, J. Eatock, S. McClean, D. Liu, X. Liu and T. Young, Modeling throughput of emergency departments via time series: an expectation maximization algorithm, *ACM Transactions on Management Information Systems*, vol. 4, no. 4, pp. 16, 2013.
- [25] N. Zeng, Z. Wang, H. Zhang and F. E. Alsaadi, A novel switching delayed PSO algorithm for estimating unknown parameters of lateral flow immunoassay, *Cognitive Computation*, vol. 8, no. 2, pp. 143–152, 2016.
- [26] N. Zeng, Z. Wang, H. Zhang, W. Liu and F. E. Alsaadi, Deep belief networks for quantitative analysis of a gold immunochromatographic strip, *Cognitive Computation*, vol. 8, no. 4, pp. 684–692, 2016.